

① (5 marks)

Find the value of $\int_{t=-\infty}^{+1} e^t \sin\left(t + \frac{\pi}{4}\right) dt$.

② (5 marks)

Suppose $x(t) = \begin{cases} t, & 0 \leq t \leq 1, \\ 0, & \text{otherwise} \end{cases}$

Sketch and label carefully $y(t) = x(3-2t)$.

lecture-6I - Best Approximation

Scenario: T -periodic signal x is given.
Integer N is also given.

Find $c_n, \overbrace{n=0, \pm 1, \dots, \pm N}^{+N}$
such that $\underbrace{\sum_{n=-N}^{+N} c_n e^{jn\omega t}}_{\text{approximates}}$
 $\underbrace{x(t)}_{\text{as closely as possible. Here } \omega = \frac{2\pi}{T}}$

↓
minimise the energy of the error signal

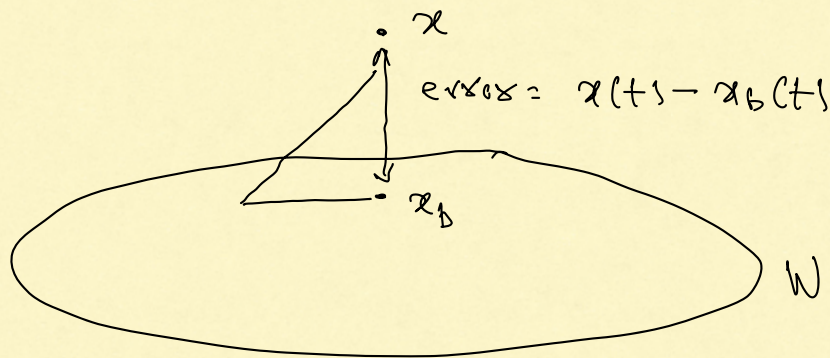
Define $\mathcal{W}$ to be the collection of all signals of the form $\sum_{n=-N}^{+N} v_n e^{jn\omega t}$, where $\underbrace{v_0, v_1, \dots, v_N}_{v_{-1}, \dots, v_{-N}} \in \mathbb{C}$

Recall: W is a vector space of dimension $2N+1$.

Idea: Claim that the best approximation to x is

$$x_b(t) = \sum_{-N}^{+N} c_n e^{jn\omega t},$$

$$c_n = \frac{1}{T} (x, e^{jn\omega t})$$



$$\text{error } e(t) = x(t) - x_b(t).$$

Step 1: With x_b and error $e(t)$ as defined above we now show that $e(t)$ is orthogonal to every signal in W .

Pick a signal $y \in W$.

$\Rightarrow$ y is a linear combination of $e^{jn\omega t}$, $n = 0, \pm 1, \dots, \pm N$.

$\Rightarrow \exists$ scalars $\gamma_0, \gamma_{\pm 1}, \dots, \gamma_{\pm N}$ such that

$$y(t) = \sum_{-N}^{+N} \gamma_n e^{jn\omega t}.$$

Inner product of y with error:

$$(x - x_b, y) = (x - x_b, \sum r_n e^{jn\omega t})$$

$$= \sum_{n=-N}^{+N} (x - x_b, r_n e^{jn\omega t})$$

$$= \sum_{n=-N}^{+N} r_n^* (x - x_b, e^{jn\omega t})$$

$$= \sum_{n=-N}^{+N} r_n^* \left[(x, e^{jn\omega t}) - (x_b, e^{jn\omega t}) \right]$$

$$= \sum_{n=-N}^{+N} r_n^* \left[T c_n - \left(\sum_{k=-N}^{+N} c_k e^{jk\omega t}, e^{jn\omega t} \right) \right]$$

$$= \sum_{n=-N}^{+N} r_n^* T c_n - \sum_{n=-N}^{+N} r_n^* \left(\sum_{k=-N}^{+N} c_k e^{jk\omega t}, e^{jn\omega t} \right)$$

$$= \quad " \quad - \sum_{n=-N}^{+N} \sum_{k=-N}^{+N} r_n^* c_k \cdot (e^{jk\omega t}, e^{jn\omega t})$$

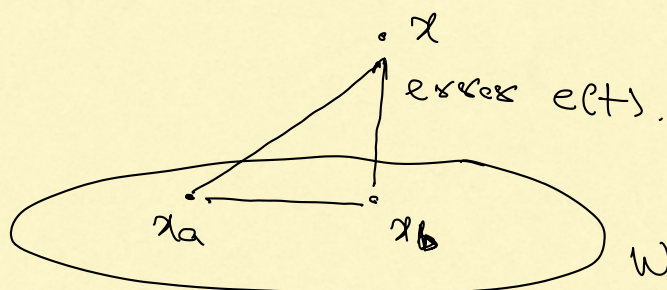
↪ $\delta_{k,n} \cdot T$

$$= \sum_n T r_n^* c_n - \sum_n \sum_k \delta_{k,n} T \cdot r_n^* c_k$$

$$= \sum_n T r_n^* c_n - \sum_n T r_n^* c_n$$

$$= 0$$

Step 2 :



Let x_a be any signal from W .

Need to show: $\|x - x_a\|^2 \geq \|x - x_b\|^2$

Main idea: $x_a \in W$, $x_b \in W$ & W is a vector space.

$$x_a - x_b, x_b - x_a \in W$$

$$\|x - x_a\|^2 = \underbrace{\|x - x_b\|^2}_{\text{error}} + \underbrace{\|x_b - x_a\|^2}_{\in W}$$

$$= \|x - x_b\|^2 + \underbrace{\|x_b - x_a\|^2}_{\geq 0} \quad \left(\begin{array}{l} \text{since error} \\ \text{is orthogonal} \\ \text{to } W \end{array} \right)$$

$$\geq \|x - x_b\|^2$$

II. Definitions / notations.

Defn: Given a T -periodic signal x the number

$$c_n = \frac{1}{T} \langle x, e^{jn\omega t} \rangle$$

$$= \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt, \quad \omega = 2\pi/T$$

is called the n^{th} Fourier series coefficient of x .

Defn: With c_n defined as above and $N > 0$ any integer

$$S_N(t) = \sum_{n=-N}^{+N} c_n e^{jn\omega t}$$

is called the N^{th} partial sum of Fourier series of x .

Main Results:

Fourier Analysis: An Introduction } Chap. 3
By Stein & Shakarchi

Theorem: Let x be Riemann integrable.

1. Mean-Square Convergence

$$\|x - S_N\|^2 \rightarrow 0 \text{ as } N \rightarrow \infty$$

2. Parseval's Theorem:

$$\|x\|^2 = \sum_{-\infty}^{+\infty} |c_n|^2$$

3. Riemann-Lebesgue Lemma:

$$c_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$c_n \rightarrow 0 \text{ as } n \rightarrow -\infty$$

4. $x \mapsto c_n$ $y \mapsto d_n$ (Fourier series coefficients)

$$(x, y) = \sum_{-\infty}^{+\infty} c_n d_n^*$$